

5. Reconciliation

Reconciliation is a critical financial process used to ensure **consistency between different financial systems** that operate asynchronously, such as banks, card networks, and payment processors.

5.1 What is Reconciliation?

Reconciliation is the process of **comparing transaction records between two systems** to ensure that:

- Every transaction recorded in one system exists in the other
- Amounts match exactly
- No transactions are missing, duplicated, or inconsistent

Since financial systems (banks, card networks, payment gateways) communicate asynchronously, discrepancies can naturally occur. Reconciliation acts as a **corrective mechanism**.

5.2 Why Reconciliation is Needed

Banks and payment systems do not operate in real time synchronization. Instead:

- Transactions are processed in batches
- Network delays may occur
- Failures may happen at intermediate steps
- Some transactions may succeed in one system but fail to reflect in another

Therefore, reconciliation is necessary to:

- Detect mismatched transactions
- Identify missing or duplicate entries
- Correct financial inconsistencies
- Ensure regulatory compliance and auditability

5.3 Bank-Level Reconciliation

At the banking level:

- Banks periodically (e.g., daily or every few days) compare their **ledger logs**
- They exchange transaction records with counterparties
- Any mismatch triggers:
 - Investigation
 - Adjustment entries (credit/debit corrections)
 - Rollback or reprocessing of transactions

This ensures that both sides of the financial ecosystem remain consistent over time.

5.4 Stripe-Level Reconciliation

As a payment processor, Stripe must also perform reconciliation, especially for:

- **Merchant payouts**
- **Bank settlement records**
- **Card network settlement reports**
- **Internal ledger consistency**

Stripe compares:

- Internal ledger records (what Stripe believes happened)
- vs
- External bank statements (what actually settled)

5.5 Reconciliation in Payout Flow

A key use case is **merchant payouts**:

1. Stripe records successful payments in its internal ledger
2. Stripe initiates payouts to merchant bank accounts
3. After settlement, Stripe receives external confirmation from banks
4. Reconciliation ensures:
 - Paid amount = settled amount
 - No missing or duplicate payouts
 - Fees are correctly applied

5.6 What Happens When Mismatch Occurs?

If discrepancies are detected:

- Missing transaction → reprocessing or manual correction
- Extra transaction → reversal or adjustment entry
- Partial mismatch → investigation and correction entry in ledger

Importantly:

- Nothing is deleted
- Corrections are done via **adjustment entries in ledger**

5.7 Key Insight

Reconciliation is not about preventing errors—it is about **detecting and correcting inevitable inconsistencies** in distributed financial systems.

5.8 Summary

- Reconciliation ensures consistency across distributed financial systems
- It is required because systems are asynchronous
- Banks reconcile with each other periodically
- Stripe reconciles internal ledger with external bank and network records
- Corrections are handled via ledger adjustments, not deletions